



Solution Set

Question 1

Consider a collateralized forward, FRA, or vanilla swap desk position. Define the state variables (discount factors, forwards, accrual factors, and payoff dates) and derive the present-value and par-rate relation. State the measure, filtration, and tradability assumptions.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Answer key

Define the tradable state variables first: discount factors, forwards, accrual factors, and payoff dates. The payoff is then written at the relevant horizon and valued under the pricing measure.

The derivation uses the economic relation for present-value and par-rate relation. If a hedge or replication argument is used, the stochastic exposure is neutralized locally and the remaining cash account earns the risk-free or collateral rate.

The final expression must agree with the payoff at maturity and with the obvious limiting cases.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$d\Pi = dV - \Delta dS, \quad d\Pi = r\Pi dt$$

Question 2

For a collateralized forward, FRA, or vanilla swap desk position, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Answer key

The pricing measure is chosen so discounted tradable prices are martingales under the chosen numeraire.

The Radon-Nikodym density reweights probabilities; it does not change the contractual payoff. In Brownian models, Girsanov changes the drift and leaves the quadratic variation unchanged.

A useful check is that the density has expectation one and that the resulting discounted price process has zero drift.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$Z_t = \frac{dQ}{dP} \Big|_{\mathcal{F}_t}, \quad E^Q[X \mid \mathcal{F}_t] = \frac{1}{Z_t} E^P[Z_T X \mid \mathcal{F}_t]$$

Question 3

Identify the discounted tradable or density-adjusted process in a collateralized forward, FRA, or vanilla swap desk position. Prove the martingale condition and name one modelling choice that would break it.

Answer key

The martingale candidate is the discounted tradable price, or the density-adjusted process after a measure change.

Condition on the earlier information set and verify that the conditional expectation equals the earlier value. Any remaining predictable drift indicates a missing funding, dividend, numeraire, or model adjustment.

For this topic, the practical issue is double-counting discounting or applying a par-rate formula to non-par cashflows.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$E[M_t \mid \mathcal{F}_s] = M_s, \quad 0 \leq s < t$$

Question 4

Using a semiannual five-year swap curve with spot zero rates between 3.3 percent and 4.2 percent, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Answer key

Use the stated market inputs: a semiannual five-year swap curve with spot zero rates between 3.3 percent and 4.2 percent. Compute future state values first, then take the correct conditional average, then discount.

Keep a calculation table with states, probabilities, payoffs, discount factors, and final present values. This makes unit mistakes visible.

The answer should include a sign and magnitude check, not only a final number.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$V_t = e^{-r\Delta t} (qV_u + (1-q)V_d)$$

Question 5

For a collateralized forward, FRA, or vanilla swap desk position, compute and interpret PV01, forward-rate delta, and convexity correction. Explain the hedge sign and the unit of each sensitivity.

Answer key

The relevant exposure is PV01, forward-rate delta, and convexity correction. Compute the first-order term, then add convexity or cross terms only when the state variables require them.

A hedge has the opposite sign of the exposure. Delta is per unit of underlying, DV01 is per basis point, and vega is per volatility point or percent depending on convention.

After hedging, report the residual risk that remains.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \nu d\sigma$$

Question 6

Suppose the desk observes deposit, futures, and swap par-rate quotes. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Answer key

The calibration instruments are deposit, futures, and swap par-rate quotes. Write a weighted least-squares objective using market quotes and model quotes.

Weights should reflect liquidity, bid-ask width, and quote reliability. Parameter constraints should prevent impossible prices or unstable extrapolation.

After calibration, inspect residuals and check whether each parameter has a distinct effect on the quotes.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$\hat{\theta} = \arg \min_{\theta} \sum_i w_i (M_i(\theta) - Q_i^{\text{mkt}})^2$$

Question 7

State the terminal and boundary conditions for a collateralized forward, FRA, or vanilla swap desk position. Explain how linear payoff maturity values and discount-factor limits prevents economically impossible prices.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Answer key

The terminal condition comes from the payoff. Boundary conditions come from economic limits such as zero spot, very large spot, barrier touch, exercise, or discount-factor positivity.

For this product, the key boundary idea is linear payoff maturity values and discount-factor limits.

A boundary condition is wrong if it permits arbitrage, negative values for non-negative payoffs, or a price that violates obvious monotonicity.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$V(T, S) = g(S), \quad V(t, S_{\text{boundary}}) = \text{economic boundary value}$$

Question 8

Design a Monte Carlo or lattice estimator for a collateralized forward, FRA, or vanilla swap desk position. Specify scenario shocks to the curve and forward level, the estimator, the discounting rule, and the standard error.

Answer key

Simulate scenario shocks to the curve and forward level. For each path, compute the payoff using only information available along that path.

The estimator is the discounted sample average. Report standard error and rerun with larger path counts or finer time steps to separate sampling error from discretization error.

For path-dependent products, store the running state variable explicitly.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$\hat{V}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \widehat{\text{SE}} = e^{-rT} \frac{s_X}{\sqrt{N}}$$

Question 9

A junior quant implements a collateralized forward, FRA, or vanilla swap desk position and gets a stable but wrong answer. Explain how double-counting discounting or applying a par-rate formula to non-par cashflows creates the error and design a diagnostic test.

Answer key

The main implementation risk is double-counting discounting or applying a par-rate formula to non-par cashflows. This can produce stable but wrong results because the error is systematic.

A good diagnostic compares the implementation with a limiting case or benchmark where the direction of the answer is known.

Change one input at a time and verify monotonicity, units, and convergence.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

$$\text{error}(N, \Delta t) = \widehat{V}_{N, \Delta t} - V_{\text{benchmark}}$$

Question 10

Write an NumPy algorithm for a collateralized forward, FRA, or vanilla swap desk position. Include inputs, outputs, diagnostics, and one benchmark case.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Answer key

Create clear arrays for inputs, states, probabilities, discount factors, and payoffs. Check dimensions before combining them.

Return both the main value and diagnostics such as convergence, residuals, standard error, or benchmark difference.

The routine is acceptable only if the result can be reproduced from the displayed assumptions and data.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

$$\text{tree cost} = O(n^2), \quad \text{simulation cost} = O(NM)$$

Question 11

Starting from the payoff definition of a collateralized forward, FRA, or vanilla swap desk position, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Answer key

Define the tradable state variables first: discount factors, forwards, accrual factors, and payoff dates. The payoff is then written at the relevant horizon and valued under the pricing measure.

The derivation uses the economic relation for present-value and par-rate relation. If a hedge or replication argument is used, the stochastic exposure is neutralized locally and the remaining cash account earns the risk-free or collateral rate.

The final expression must agree with the payoff at maturity and with the obvious limiting cases.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

$$d\Pi = dV - \Delta dS, \quad d\Pi = r\Pi dt$$

Question 12

Create a stressed-market version of a semiannual five-year swap curve with spot zero rates between 3.3 percent and 4.2 percent. Recompute the value or risk metric and explain which term dominates the change.

Answer key

Use the stated market inputs: a semiannual five-year swap curve with spot zero rates between 3.3 percent and 4.2 percent. Compute future state values first, then take the correct conditional average, then discount.

Keep a calculation table with states, probabilities, payoffs, discount factors, and final present values. This makes unit mistakes visible.

The answer should include a sign and magnitude check, not only a final number.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$V_t = e^{-r\Delta t} (qV_u + (1-q)V_d)$$

Question 13

Use a second-order Taylor approximation for a collateralized forward, FRA, or vanilla swap desk position. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Answer key

The relevant exposure is PV01, forward-rate delta, and convexity correction. Compute the first-order term, then add convexity or cross terms only when the state variables require them.

A hedge has the opposite sign of the exposure. Delta is per unit of underlying, DV01 is per basis point, and vega is per volatility point or percent depending on convention.

After hedging, report the residual risk that remains.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \nu d\sigma$$

Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a collateralized forward, FRA, or vanilla swap desk position. Identify what becomes a martingale after the change.

Answer key

The pricing measure is chosen so discounted tradable prices are martingales under the chosen numeraire.

The Radon-Nikodym density reweights probabilities; it does not change the contractual payoff. In Brownian models, Girsanov changes the drift and leaves the quadratic variation unchanged.

A useful check is that the density has expectation one and that the resulting discounted price process has zero drift.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$Z_t = \frac{dQ}{dP} \Big|_{\mathcal{F}_t}, \quad E^Q[X \mid \mathcal{F}_t] = \frac{1}{Z_t} E^P[Z_T X \mid \mathcal{F}_t]$$

Question 15

Give a proof-style answer showing when the wealth process for a collateralized forward, FRA, or vanilla swap desk position is self-financing. Explain why predictable trading strategies matter.

Answer key

The martingale candidate is the discounted tradable price, or the density-adjusted process after a measure change.

Condition on the earlier information set and verify that the conditional expectation equals the earlier value. Any remaining predictable drift indicates a missing funding, dividend, numeraire, or model adjustment.

For this topic, the practical issue is double-counting discounting or applying a par-rate formula to non-par cashflows.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \quad 0 \leq s < t$$

Question 16

Market quotes for deposit, futures, and swap par-rate quotes are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Answer key

The calibration instruments are deposit, futures, and swap par-rate quotes. Write a weighted least-squares objective using quotes and model quotes.

Weights should reflect liquidity, bid-ask width, and quote reliability. Parameter constraints should prevent impossible prices or unstable extrapolation.

After calibration, inspect residuals and check whether each parameter has a distinct effect on the quotes.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$\hat{\theta} = \arg \min_{\theta} \sum_i w_i (M_i(\theta) - Q_i^{\text{mkt}})^2$$

Question 17

For a collateralized forward, FRA, or vanilla swap desk position, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Answer key

The terminal condition comes from the payoff. Boundary conditions come from economic limits such as zero spot, very large spot, barrier touch, exercise, or discount-factor positivity.

For this product, the key boundary idea is linear payoff maturity values and discount-factor limits.

A boundary condition is wrong if it permits arbitrage, negative values for non-negative payoffs, or a price that violates obvious monotonicity.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$
$$V(T, S) = g(S), \quad V(t, S_{\text{boundary}}) = \text{economic boundary value}$$

Question 18

Design a convergence study for scenario shocks to the curve and forward level. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Answer key

Simulate scenario shocks to the curve and forward level. For each path, compute the payoff using only information available along that path.

The estimator is the discounted sample average. Report standard error and rerun with larger path counts or finer time steps to separate sampling error from discretization error.

For path-dependent products, store the running state variable explicitly.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

$$\widehat{V}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \widehat{SE} = e^{-rT} \frac{s_X}{\sqrt{N}}$$

Question 19

Find one way that double-counting discounting or applying a par-rate formula to non-par cashflows could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Answer key

The main implementation risk is double-counting discounting or applying a par-rate formula to non-par cashflows. This can produce stable but wrong results because the error is systematic.

A good diagnostic compares the implementation with a limiting case or benchmark where the direction of the answer is known.

Change one input at a time and verify monotonicity, units, and convergence.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

$$\text{error}(N, \Delta t) = \widehat{V}_{N, \Delta t} - V_{\text{benchmark}}$$

Question 20

Explain a collateralized forward, FRA, or vanilla swap desk position and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.

Answer key

Create clear arrays for inputs, states, probabilities, discount factors, and payoffs. Check dimensions before combining them.

Return both the main value and diagnostics such as convergence, residuals, standard error, or benchmark difference.

The routine is acceptable only if the result can be reproduced from the displayed assumptions and data.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

$$\text{tree cost} = O(n^2), \quad \text{simulation cost} = O(NM)$$